

Deimonas Suchanka

Validation & Quality Engineer | The Hague, Netherlands

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Analytical Systems Engineer who solves complex electro-mechanical problems through data-driven root cause analysis. My work has consistently involved engineering new processes and technical frameworks that deliver significant, quantifiable improvements in system performance and reliability.

KEY ENGINEERING PROJECTS

- **Root Cause Analysis: Pekos Clamping System** *See Appendix A*
Diagnosed a critical gain margin instability through the first-ever data-driven analysis in the workshop's history, deploying a validated control solution that improved force consistency by over 350%.
- **Architecture: Tiered QA Validation Framework** *See Appendix B*
Architected a new, data-driven QA framework and a lean Standard Operating Procedure (SOP) to formalize knowledge capture and provide objective quality evidence in alignment with ISO 9001 principles.
- **System: "Docs-as-Code" Technical Documentation** *See Appendix C*
Replaced a chaotic, file-based process by architecting and implementing a version-controlled workflow (Git, LaTeX, MkDocs) that now serves as the single source of truth for all technical manuals.

PROFESSIONAL EXPERIENCE

Ventil Test Equipment BV Rijswijk, The Netherlands
Service Engineer (Systems & Process Focus) November 2022 – Present

- Improved production consistency by +350% by executing a data-driven root cause analysis on a critical electro-mechanical clamping system.
- Architected a new, version-controlled "docs-as-code" workflow using Git & LaTeX where no formal system existed.
- Designed a tiered QA validation framework to formalize knowledge capture and align workshop processes with ISO 9001 principles.

Fast Fluid Management Sassenheim, Netherlands
Assembly Technician May 2022 – October 2022

- Assembled, tested, and troubleshooted complex mechanical and electrical dispensing systems independently.
- Performed quality control checks to ensure all machines met operational standards before shipping.

Systems Analysis: Root Cause Analysis (RCA), Control Systems (Gain Margin), Data-Driven Diagnosis
Process Architecture: QA Framework Design, ISO 9001 Principles, Standard Operating Procedure (SOP) Creation
Technical Documentation: "Docs-as-Code," LaTeX, MkDocs, Git/GitHub, P&ID Interpretation (progeCAD)

CERTIFICATIONS & CLEARANCES

VOL-VCA (Safety for Operational Supervisors)
EU/NATO Citizen, Lithuanian Armed Forces Veteran (2020-2021), Eligible for Security Clearance

EDUCATION

University of Amsterdam Amsterdam, Netherlands
Coursework toward Bachelor of Arts, European Studies 2021 – 2022

- Developed strong academic writing and research skills.

Root Cause Analysis: Pekos Clamping System

Appendix: Case Study A

+350% Improvement in Force Consistency

The first-ever data-driven analysis of a critical production system, diagnosing a core instability and deploying a validated control solution that became the blueprint for future QA.

THE PROBLEM

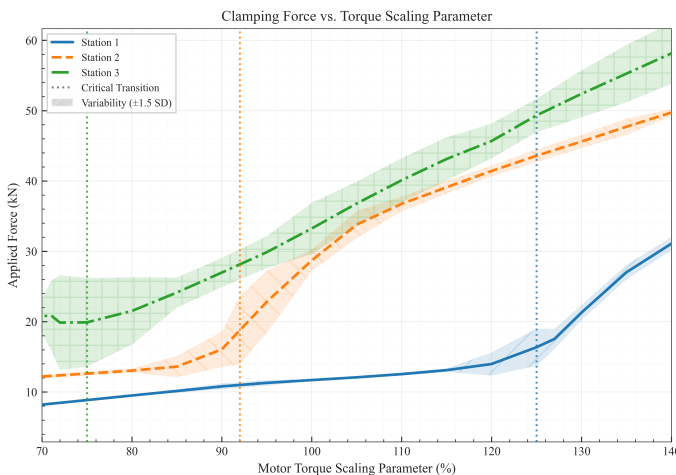
The Pekos clamping system, a key production asset, exhibited severe force inconsistency, leading to quality issues and line stoppages. Previous troubleshooting was based on intuition and had failed to find a root cause. The objective was to diagnose and resolve the instability with empirical data.

METHODOLOGY

- Data Acquisition:** Deployed sensors to capture real-time performance data (pressure, position, force) from the hydraulic system and control board, establishing a quantitative baseline.
- Data Analysis:** Analyzed the time-series data using Python (NumPy, Matplotlib), which revealed a critical gain margin instability in the PID controller's feedback loop causing oscillations under load.
- Solution Development:** Modeled the system dynamics and engineered a revised set of PID tuning parameters. Designed a validation test to compare old vs. new parameters under controlled conditions.
- Deployment Validation:** Implemented the validated control parameters on the production machine and created a new Standard Operating Procedure (SOP) for calibration, confirming the fix with post-implementation data.

DATA VISUALIZATION & IMPACT

The graph below clearly shows the instability (blue, orange) under the old parameters versus the stable, consistent force application (green) achieved with the new control solution.



DELIVERABLES

Engineering Analysis Report

An 11-page, IEEE-formatted technical report detailing the full root cause analysis, methodology, data visualization, and validated results.

QA Validation Framework

A strategic proposal and lean SOP designed to formalize knowledge capture and provide objective QA evidence.

VFD Parameter Set

A finalized master list of station-specific inverter parameters, detailing the precise values for torque detection, deceleration, and force scaling required to achieve stable operation.

Architecture: Tiered QA Validation Framework

Appendix: Case Study B

From Ad-Hoc to Standardized QA

Architected a new validation framework to move from basic functional checks to objective, data-driven quality assurance, addressing senior management's goal for improved knowledge capture.

THE PROBLEM

Existing quality checks (FAT/IAT) verified basic functionality but missed subtle performance variations. This created three core business problems: a lack of objective data for quality assurance, no formal process for capturing engineering knowledge, and a clear gap in alignment with ISO 9001 standards.

METHODOLOGY

- Foundational Analysis:** Leveraged findings from the Pekos system analysis (Appendix A) as a catalyst to demonstrate the need for a more rigorous, data-driven validation process.
- Workflow Modeling:** Analyzed the existing ad-hoc workflow and designed a new, structured process with clear decision points and escalation paths.
- Framework Architecture:** Designed a tiered (Level 1-3) system to efficiently handle everything from routine QA checks to deep-dive root cause analysis for complex issues.
- Metric & Procedure Definition:** Defined specific, data-driven performance metrics (e.g., Consistency Index) and developed a lean, actionable SOP for Level 1 validation.

FRAMEWORK ARCHITECTURE

- Level 1: Routine Validation**
A lean, standardized procedure for commissioning engineers to perform data-driven pass/fail checks on all machines, providing objective QA evidence.
- Level 2: Targeted Diagnosis**
A focused investigation, triggered by a Level 1 failure, to quickly diagnose the issue and determine a corrective action (e.g., parameter tuning, mechanical fix).
- Level 3: Deep-Dive RCA**
A full, systematic root cause analysis involving a cross-functional team for complex, systemic issues, modeled on the Pekos investigation.

DELIVERABLES

QA Framework Proposal	A formal proposal outlining the tiered (L1-L3) framework, stakeholder impact, key metrics, and implementation plan.
Level 1 Test Procedure	A lean, single-page SOP for conducting routine performance validation in the workshop.
Data Template	An Excel template for data collection and automatic calculation of the Consistency Index (CI) metric during Level 1 tests.

**Note: These documents contain proprietary information and are available to be discussed in detail during an interview.*

System: "Docs-as-Code" Technical Documentation Workflow

Appendix: Case Study C

Foundation for Automated Documentation

Independently architected a modern documentation system from the ground up, replacing a chaotic file-based process with a version-controlled, single-source-of-truth workflow.

THE PROBLEM

The existing process for creating technical manuals was non-existent. It consisted of scattered Word documents on a shared drive with no version control, inconsistent formatting, and outdated information. This led to customer complaints and significant internal inefficiency.

METHODOLOGY

- Research Strategic Pivot:** Initially researched a complex pipeline for legacy '.docx' workflows, then pivoted to a more robust and pragmatic "docs-as-code" approach from first principles.
- System Development:** Established a complete development environment using VSCode, created a comprehensive style guide, and built out reusable templates through iterative experimentation.
- Implementation Delivery:** Authored new, high-quality technical manuals using the system, including taking ownership of custom technical illustrations by editing P&ID schematics in progeCAD.

SOLUTION ARCHITECTURE

- Version Control as Foundation:** Used Git for rigorous version control, providing a complete history of changes and establishing a single source of truth for all documentation.
- Dual-Output Workflow:** Architected two parallel pipelines: a LaTeX-based workflow for superior print quality, and a Markdown-based workflow for responsive, searchable static websites (using MkDocs).
- Foundation for Automation:** Designed the entire system with a "Lego block" philosophy, creating a modular foundation to enable the future automatic generation of machine-specific manuals from a given configuration.

DELIVERABLES

Documentation Workflow	A complete, version-controlled system for generating both print (LaTeX) and web (Markdown/MkDocs) docs from a structured process.
Style Guide & Templates	A comprehensive style guide and set of boilerplate templates to ensure all future documentation is consistent and professional.
Software Manual Prototype	A functional static website and self-contained '.exe' (via Tauri 2.0) for the CRS10 software manual, approved by the lead developer.

*Note: The system architecture and templates are proprietary and are available to be discussed in detail during an interview.